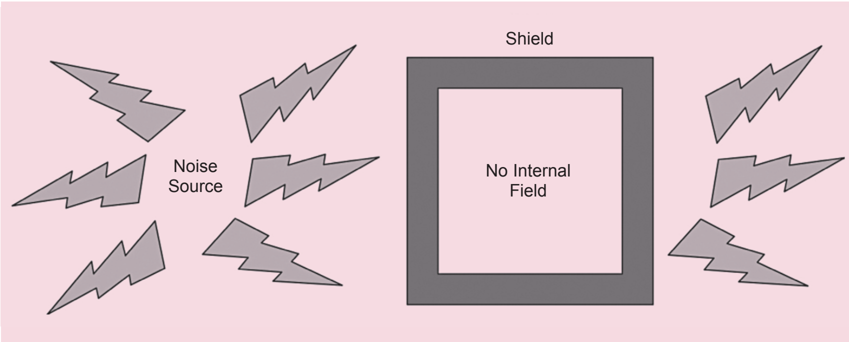

Fig. 5: Shield prevents radiation from entering the enclosure

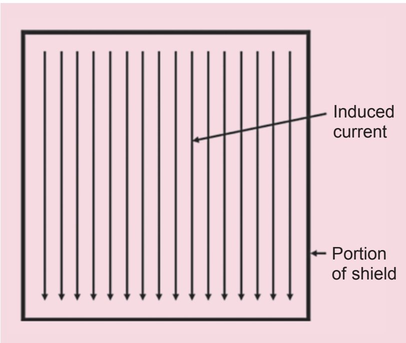

Fig. 6: How induced current flows when there are no apertures

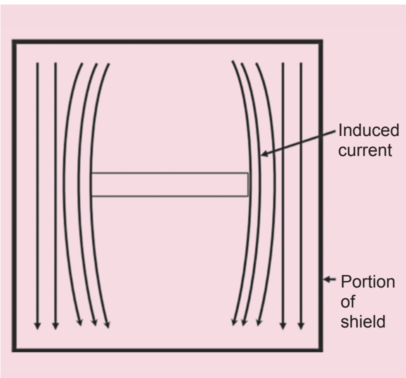

Fig. 7: How current flow gets disturbed when there is an aperture

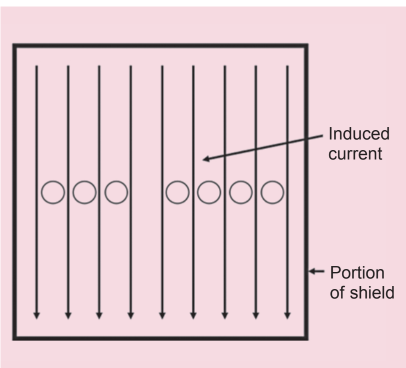

Fig. 8: Ventilation through a cluster of holes allows the induced current to flow undisturbed

Class A allows higher radiation as industrial environment is very noisy. However, Class B has much lower (stricter) radiation limit and most consumer goods need to meet this standard.

A note of caution. *Designers think EMC and EMI problems are due to enclosures alone. This is a myth every designer should be aware of. For a successful compliance, the entire product has to be designed to meet the standards. PCB assemblies (PCBAs) are the major culprits and these need careful EMC/EMI design in addition to enclosures.*

Solutions for radiated and conducted emission vary in implementation. Primary details needed are the frequencies at which the electronics of the system operates and the RF carrier frequency if the product is a radio equipment. Basic square waves tend to contain harmonics, which contribute to the radiation in a big way. A solution to radiated emission is reduction in emissions from the PCBA, which basically depends on the circuit design. Experience shows high-speed digital circuits are the main sources for radiation as these have square waves with small rise time.

Circuit design for EMC/EMI reduction is a vast topic and is not being covered here as our primary focus is on electronic packaging. However, radiations from PCBAs can only be reduced and cannot be eliminated fully, and that job has to be done by the enclosure and filters.

The process of containing RF radiation is called shielding. Metals are the best shields to contain radiated emission. So, in case of a product with plastic enclosure RF radiation containment can be a big issue. Even metal enclosures having imperfect joints, seams, or ventilation holes can be a source of RF radiation. In case of fastening mechanism of enclosures like rivets and simple screws, the space between two screws/rivets can also become a source of radiation. This is the reason why designers have to understand the compliance needs up front (SCoPE framework) to decide on the enclosure joining/fastening mechanisms.

Before getting further into the solutions for radiated and conducted emission let us understand the mechanism of shielding. Fig. 4 shows an electronics system inside a shielded enclosure with no external field. Due to the shielding no radiation escapes (radiated emission prevention). Fig. 5 shows the exact opposite where the shield prevents the radiation from entering and the system inside the enclosure is not impacted (radiated emission susceptibility) by the external radiation.

Metals are the best shielding materials due to their conduction property and therefore metal enclosures are found good for EMC/EMI protection. Plastics are bad for shielding as RF waves pass through plastics. Shielding effectiveness of a metal depends on its thickness and the frequency of RF emission. But, as a rule of thumb, mild steel has good shielding characteristics as compared to aluminium.

As we saw in Fig. 4 and Fig. 5, an ideal shield would be a tightly closed box for best performance. However, in real life, any enclosure will have seams (due to sheet-metal design) and openings for displays, switches, and cables. From a radiation perspective these are called apertures. Apertures in an enclosure lead to leakages of RF radiation. The best EMC/EMI performance is achieved when these radiation leakages are reduced to a low level.

Normally, noise fields (created by RF radiation) induce currents in the shields and these currents induce additional fields in the shield. These new fields cancel the original fields in some spaces. For this cancellation to occur, the shield currents must be allowed to flow without any hindrance. Fig. 6 shows how the induced